Hierarichal Clustering Implementation

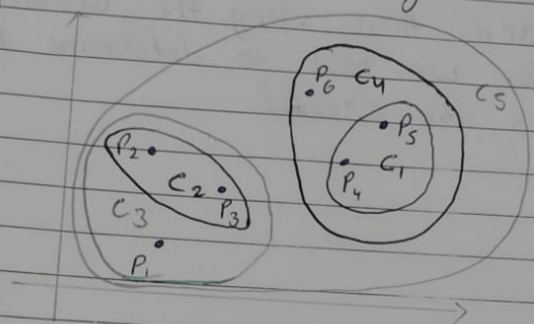
September 22, 2025

1 Hierarchal Clustering Implementation(Agglomerative Clustering)

[1]:
```python
from IPython.display import Image, display
display(Image(filename='theory.jpg'))
```

Hierarichal clustering

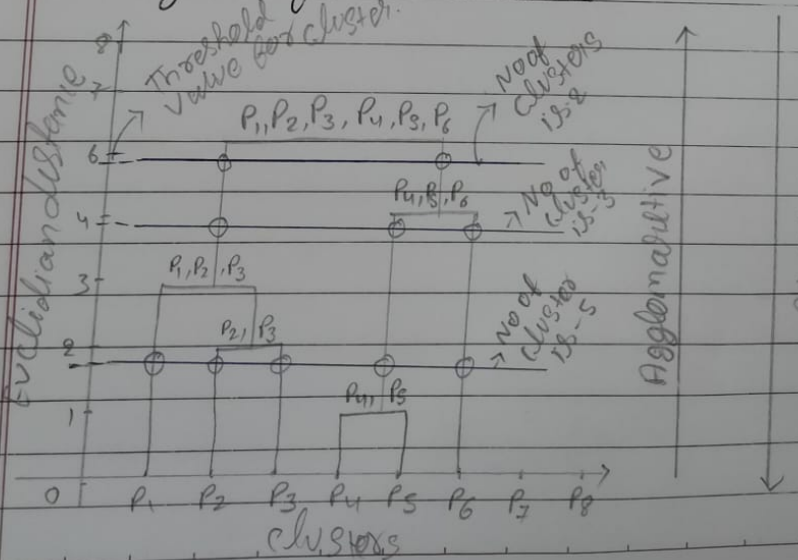
① Agglomerative → combining
② Divisive → Dividing



Steps

(i) For initially each points are consider as a seperate cluster
(ii) Find the nearest points and create a new cluster
(iii) Keep on doing the same step-1 and step-2 until we get a single cluster.

Dendogram for the clusters.



[2]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
from sklearn import datasets
```

[4]:
```python
iris = datasets.load_iris()
iris_data = pd.DataFrame(iris.data)
```

[6]:
```python
iris_data.columns = iris.feature_names
```

[8]:
```python
iris_data.head()
```

[8]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

[9]:
```python
from sklearn.preprocessing import StandardScaler
scalar = StandardScaler()
X_scaled = scalar.fit_transform(iris_data)
```

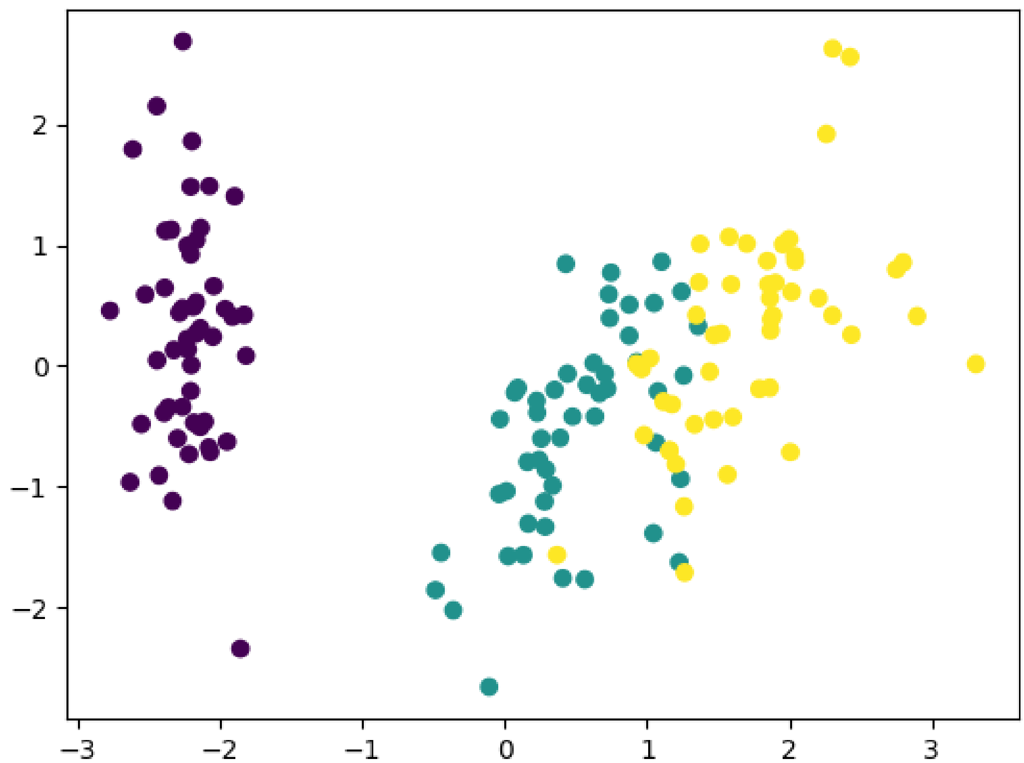
1.1 Apply PCA

[12]:
```python
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
```

[13]:
```python
pca_scaled = pca.fit_transform(X_scaled)
```

[15]:
```python
plt.scatter(pca_scaled[:,0], pca_scaled[:,1], c = iris.target)
```

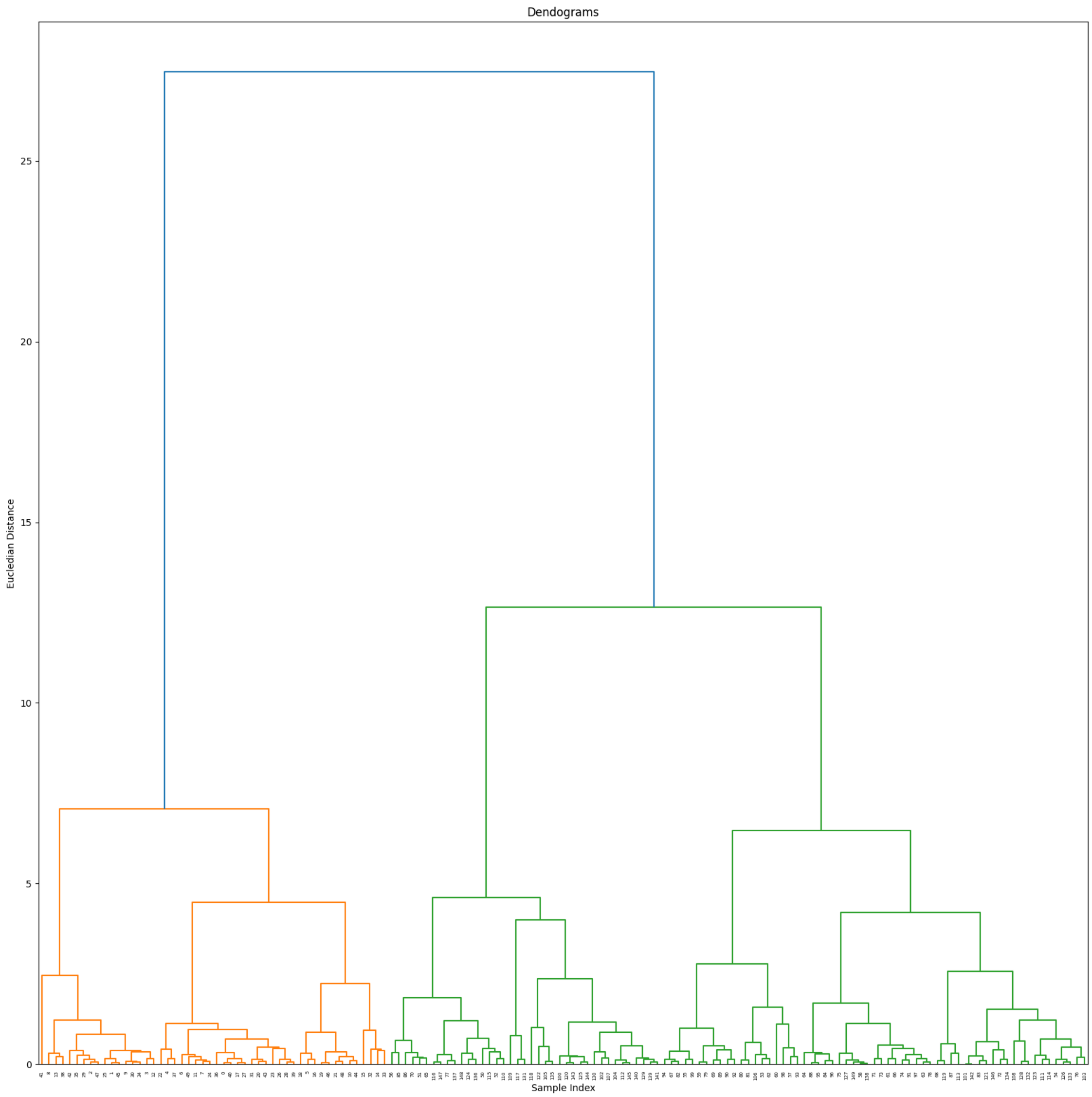
[15]:
```
<matplotlib.collections.PathCollection at 0x2f9d3da5a80>
```



1.2 Agglomerative Clustering

[18]:
```
## Constructive a dendogram
import scipy.cluster.hierarchy as sc

plt.figure(figsize=(20,20))
plt.title("Dendograms")
sc.dendrogram(sc.linkage(pca_scaled, method='ward'))
plt.xlabel("Sample Index")
plt.ylabel('Eucledian Distance')
plt.show()
```




```
[20]: from sklearn.cluster import AgglomerativeClustering
      cluster = AgglomerativeClustering(n_clusters=2, linkage='ward')
      cluster.fit(pca_scaled)
```

```
[20]: AgglomerativeClustering()
```

```
[21]: cluster.labels_
```

```
[21]: array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
             1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
             1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
             0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
             0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], dtype=int64)
```

1.2.1 Final Clusters

[23]:
```
plt.scatter(pca_scaled[:,0], pca_scaled[:,1], c = cluster.labels_)
```

[23]:
```
<matplotlib.collections.PathCollection at 0x2f9d67bf610>
```